WORKSHEET

Seasonal adjustment

Rules

Steps to deseasonalising data:

1. Find the average for each year (= *yearly average*).
2. Divide each month by the yearly average.
3. Find the average of each month (= *seasonal index* for each month).
4. Divide original data by its seasonal index.

The formula for calculating deseasonalised values is:

$$\text{Deseasonalised value} = \frac{\text{Actual value}}{\text{Seasonal index}}$$

Questions

1 Deseasonalise the data provided in each question below by completing the empty cells in each table.

a Calculate the yearly average for each row in the table (Step 1). Figures are in $1000s.

Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4	Yearly average
2011	7	4	6	3	
2012	9	5	6	4	
2013	10	5	7	3	

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00079.indd 1 16/09/15 11:20 AM

b Calculate the quarterly proportions, then the seasonal indices (SI) (Steps 2 and 3).

Year	Q1	Q2	Q3	Q4
2011	$\frac{7}{5} = 1.4$			
2012				
2013				
Total				
SI				

c Deseasonalise the data (Step 4).

Year	Q1	Q2	Q3	Q4
2011				
2012				
2013				

d Use CAS or a spreadsheet to plot both the original data and the deseasonalised time series data.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00079.indd 2 16/09/15 11:20 AM

2 a Calculate the yearly average for each row in the table (Step 1).

Year	Population February	Population May	Population August	Population November	Yearly average
2009	477	580	1228	612	
2010	478	509	1381	715	
2011	490	578	1554	653	

b Calculate the quarterly proportions, then the seasonal indices (SI) (Steps 2 and 3).

Year	Q1	Q2	Q3	Q4
2009				
2010				
2011				
Total				
SI				

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00079.indd 3 16/09/15 11:20 AM

c Deseasonalise the data (Step 4).

Year	Q1	Q2	Q3	Q4
2009				
2010				
2011				

d Use CAS or a spreadsheet to plot both the original data and the deseasonalised time series data.

© Cengage Learning Australia Pty Ltd 2015

Answers

1 a

Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4	Yearly average
2011	7	4	6	3	$\frac{7+4+6+3}{4} = 5$
2012	9	5	6	4	$\frac{9+5+6+4}{4} = 6$
2013	10	5	7	3	$\frac{10+5+7+3}{4} = 6.25$

b

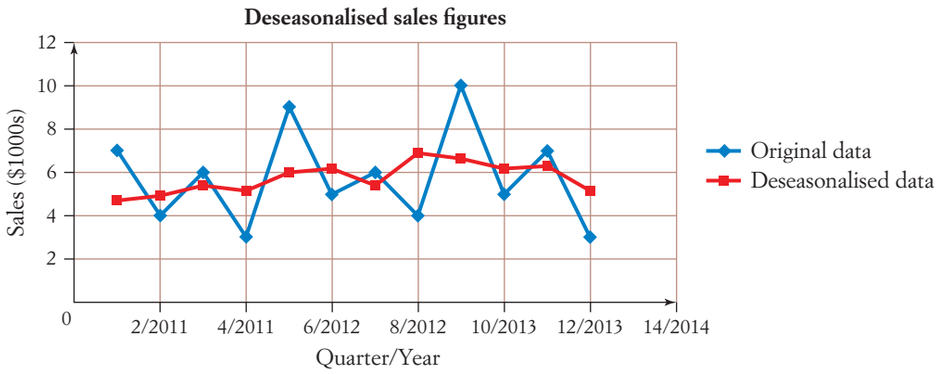
Year	Q1	Q2	Q3	Q4
2011	$\frac{7}{5} = 1.4$	$\frac{4}{5} = 0.8$	$\frac{6}{5} = 1.2$	$\frac{3}{5} = 0.6$
2012	$\frac{9}{6} = 1.5$	$\frac{5}{6} = 0.833$	$\frac{6}{6} = 1$	$\frac{4}{6} = 0.667$
2013	$\frac{10}{6.25} = 1.6$	$\frac{5}{6.25} = 0.8$	$\frac{7}{6.25} = 1.12$	$\frac{3}{6.25} = 0.48$
Total	4.5	2.433	3.32	1.747
SI	$\frac{4.5}{3} = 1.5$	$\frac{2.433}{3} = 0.811$	$\frac{3.32}{3} = 1.107$	$\frac{1.747}{3} = 0.582$

c

Year	Q1	Q2	Q3	Q4
2011	$\frac{7}{1.5} = 4.667$	$\frac{4}{0.811} = 4.932$	$\frac{6}{1.107} = 5.420$	$\frac{3}{0.582} = 5.155$
2012	$\frac{9}{1.5} = 6$	$\frac{5}{0.811} = 6.165$	$\frac{6}{1.107} = 5.420$	$\frac{4}{0.582} = 6.873$
2013	$\frac{10}{1.5} = 6.667$	$\frac{5}{0.811} = 6.165$	$\frac{7}{1.107} = 6.323$	$\frac{3}{0.582} = 5.155$

© Cengage Learning Australia Pty Ltd 2015

d



2 a

Year	Population Feb	Population May	Population August	Population November	Yearly average
2009	477	580	1228	612	$\frac{477+580+1228+612}{4} = 724.25$
2010	478	509	1381	715	$\frac{478+509+1381+715}{4} = 770.75$
2011	490	578	1554	653	$\frac{490+578+1554+653}{4} = 818.75$

b

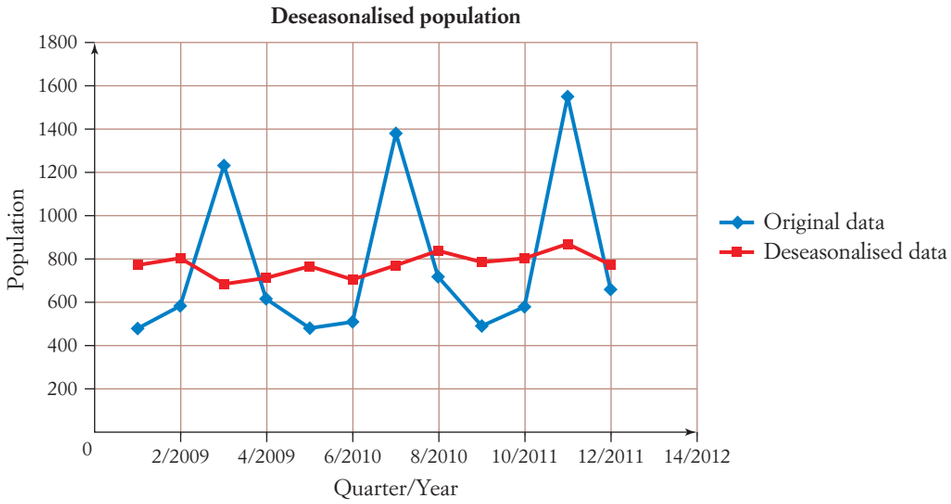
Year	Q1	Q2	Q3	Q4
2009	$\frac{477}{724.25} = 0.659$	$\frac{580}{724.25} = 0.801$	$\frac{1228}{724.25} = 1.696$	$\frac{612}{724.25} = 0.845$
2010	$\frac{478}{770.75} = 0.620$	$\frac{509}{770.75} = 0.660$	$\frac{1381}{770.75} = 1.792$	$\frac{715}{770.75} = 0.928$
2011	$\frac{490}{818.75} = 0.598$	$\frac{578}{818.75} = 0.706$	$\frac{1554}{818.75} = 1.898$	$\frac{653}{818.75} = 0.798$
Total	1.877	2.167	5.386	2.571
SI	0.626	0.722	1.795	0.857

© Cengage Learning Australia Pty Ltd 2015

c

Year	Q1	Q2	Q3	Q4
2009	$\frac{477}{0.626} = 761.98$	$\frac{580}{0.722} = 803.32$	$\frac{1228}{1.795} = 684.12$	$\frac{612}{0.857} = 714.12$
2010	$\frac{478}{0.626} = 763.58$	$\frac{509}{0.722} = 704.99$	$\frac{1381}{1.795} = 769.36$	$\frac{715}{0.857} = 834.31$
2011	$\frac{490}{0.626} = 782.75$	$\frac{578}{0.722} = 800.55$	$\frac{1554}{1.795} = 865.74$	$\frac{653}{0.857} = 761.96$

d



© Cengage Learning Australia Pty Ltd 2015